BHAVAN'S COLLEGE AUTONOMOUS, ANDHERI-WEST
PRACTICAL JOURNAL

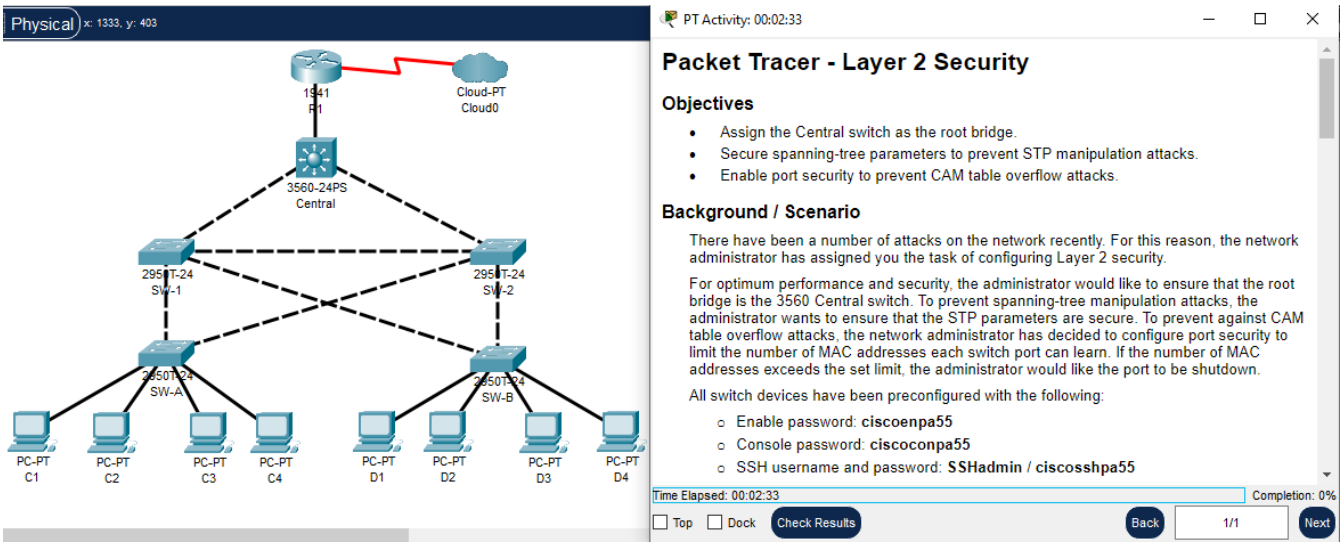
Class: TYIT **Sem: VI** **Roll No.: TYIT25** **Date:14-02-2026**

Course Name: Cybersphere and Security **Page no:**

Practical Number:- 09

Q. Layer 2 VLAN Security.

1. Download SiC All Packet Tracer Files (E-next.in) → extract it → double click on p8 Packet Tracer - Layer 2 Security → just minimize 1st page → you will get devices that connected.



2. Click on central button → CLI → enter.
   en
   password: ciscoenpa55
   show spanning-tree

```
Central>en
Password:
Central#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0009.7C61.9058
             Cost        4
             Port        25(GigabitEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     00D0.D31C.634C
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- -------------------------
Gi0/1            Root FWD 4         128.25   P2p
Gi0/2            Desg FWD 4         128.26   P2p
Fa0/1            Desg FWD 19        128.1    P2p
```

3. Click on SW-1
   en
   password: ciscoenpa55
   show spanning-tree

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```
SW-1>en
Password:
SW-1#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
             Address     0009.7C61.9058
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     0009.7C61.9058
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- ------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/23           Desg FWD 19        128.23   P2p
Fa0/24           Desg FWD 19        128.24   P2p
Gi0/1            Desg FWD 4         128.25   P2p
```

4. Click on central → CLI
   conf t
   spanning-tree vlan 1 root primary
   do show spann

```
Central#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Central(config)#spanning-tree vlan 1 root primary
Central(config)#do show spann
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     00D0.D31C.634C
             This bridge is the root
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24577  (priority 24576 sys-id-ext 1)
             Address     00D0.D31C.634C
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- ------------------------
Gi0/1            Desg FWD 4         128.25   P2p
Gi0/2            Desg FWD 4         128.26   P2p
Fa0/1            Desg FWD 19        128.1    P2p
```

5. Go on SW-1→ CLI
   conf t
   spanning-tree vlan 1 root secondary
   do show span

```
SW-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW-1(config)#spanning-tree vlan 1 root secondary
SW-1(config)#do show spann
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     00D0.D31C.634C
             Cost        4
             Port        25(GigabitEthernet0/1)
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28673  (priority 28672 sys-id-ext 1)
             Address     0009.7C61.9058
             Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  20

Interface        Role Sts Cost      Prio.Nbr Type
---------------- ---- --- --------- -------- -------------------------
Fa0/1            Desg FWD 19        128.1    P2p
Fa0/23           Desg FWD 19        128.23   P2p
Fa0/24           Desg FWD 19        128.24   P2p
Gi0/1            Root FWD 4         128.25   P2p
```

6. Go to SW-A → CLI.
   conf t
   interface range fastEthernet 0/1 - 4
   switchport mode access
   spanning-tree portfast
   spanning-tree bpduguard enable

SW-A

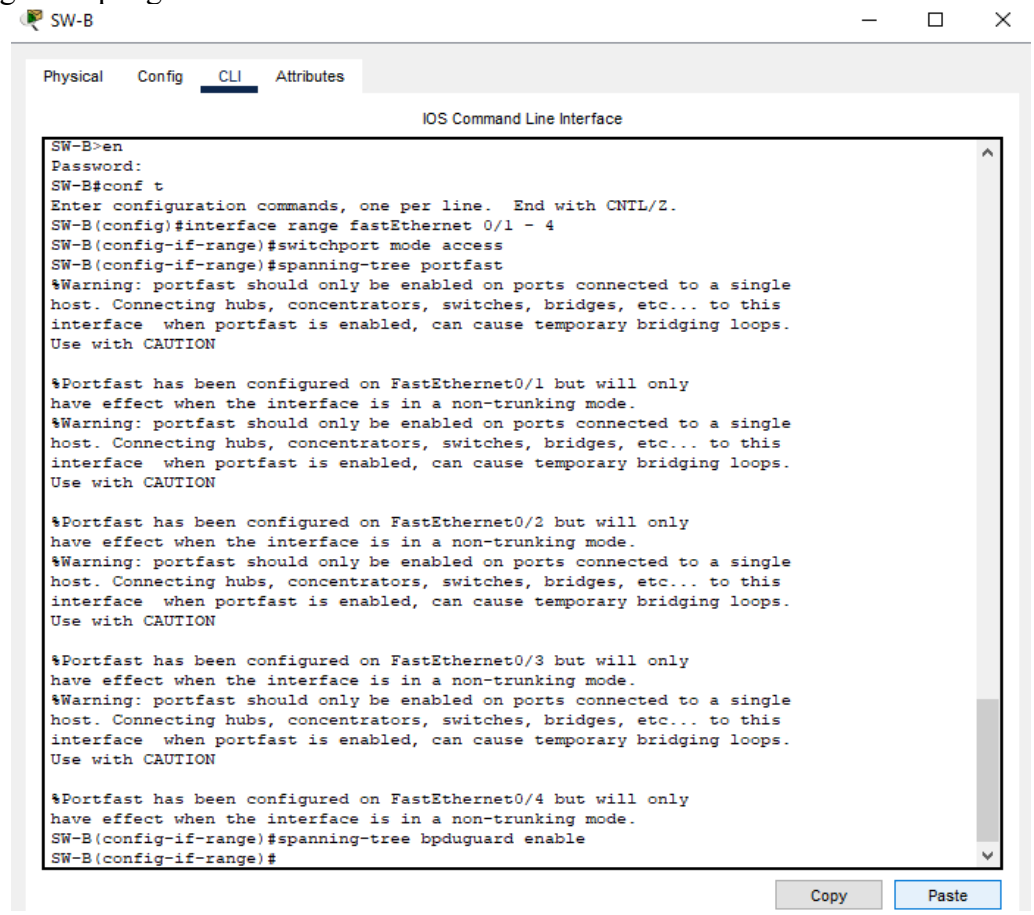
Physical Config CLI Attributes

IOS Command Line Interface

```
SW-A>en
Password:
SW-A#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW-A(config)#interface range fastEthernet 0/1 - 4
SW-A(config-if-range)#switchport mode access
SW-A(config-if-range)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/1 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/2 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/3 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface  when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on FastEthernet0/4 but will only
have effect when the interface is in a non-trunking mode.
SW-A(config-if-range)#spanning-tree bpduguard enable
SW-A(config-if-range)#
```

Copy Paste

7. Go to SW-B → CLI.
   conf t
   interface range fastEthernet 0/1 - 4
   switchport mode access
   spanning-tree portfast
   spanning-tree bpduguard enable



8. Go to SW-1 → CLI.
   interface range fastEthernet 0/23 - 24
   spanning-tree guard root

```
SW-1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW-1(config)#interface range fastEthernet 0/23 - 24
SW-1(config-if-range)#spanning-tree guard root
SW-1(config-if-range)#
```

9. Go to SW-2 → CLI.
   interface range fastEthernet 0/23 - 24
   spanning-tree guard root

```
SW-2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW-2(config)#interface range fastEthernet 0/23 - 24
SW-2(config-if-range)#spanning-tree guard root
SW-2(config-if-range)#
```

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10. Go to SW-A → CLI.

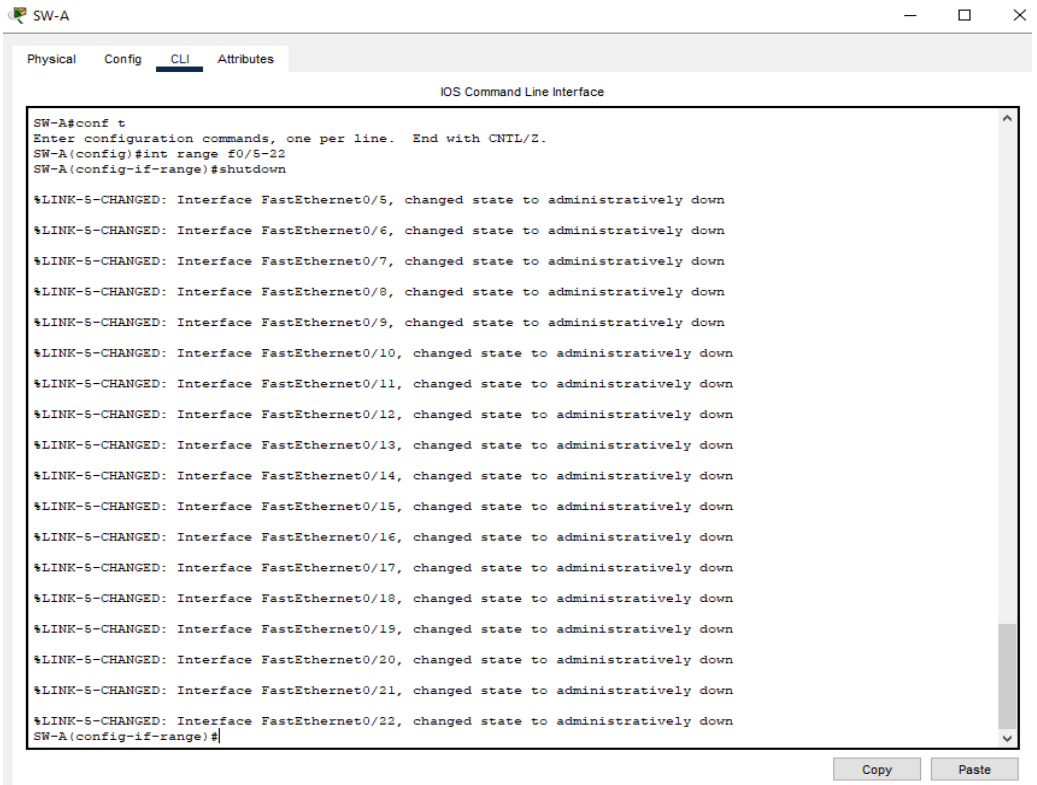
conf t
interface range fastEthernet 0/1 - 4
switchport port-security
switchport port-security maximum 2
switchport port-security mac-address sticky

```
SW-A#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW-A(config)#interface range fastEthernet 0/1 - 4
SW-A(config-if-range)#switchport port-security
SW-A(config-if-range)#switchport port-security maximum 2
SW-A(config-if-range)#switchport port-security mac-address sticky
SW-A(config-if-range)#
```

show port-security int f0/1

```
SW-A#show port-security int f0/1
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 2
Total MAC Addresses        : 0
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 0
Last Source Address:Vlan   : 0000.0000.0000:0
Security Violation Count   : 0
```

conf t
int range f0/5-22
shutdown



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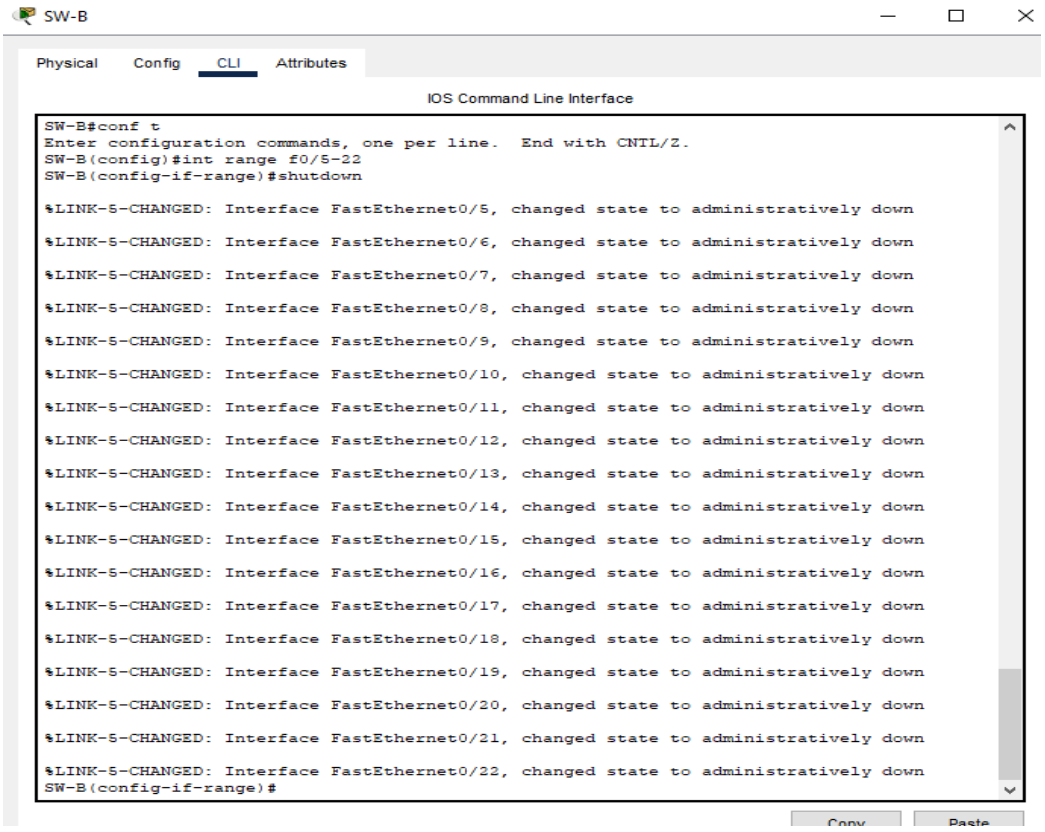
11. Same in SW-B → CLI.
    conf t
    interface range fastEthernet 0/1 - 4
    switchport port-security
    switchport port-security maximum 2
    switchport port-security mac-address sticky

```
SW-B(config)#interface range fastEthernet 0/1 - 4
SW-B(config-if-range)#switchport port-security
SW-B(config-if-range)#switchport port-security maximum 2
SW-B(config-if-range)#switchport port-security mac-address sticky
```

show port-security int f0/1

```
SW-B#show port-security int f0/1
Port Security              : Disabled
Port Status                : Secure-down
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 0
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 0
Last Source Address:Vlan   : 0000.0000.0000:0
Security Violation Count   : 0
```

conf t
int range f0/5-22
shutdown



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12. Output:

PT Activity: 01:01:12

Packet Tracer - Layer 2 Security

Objectives

- Assign the Central switch as the root bridge.
- Secure spanning-tree parameters to prevent STP manipulation attacks.
- Enable port security to prevent CAM table overflow attacks.

Background / Scenario

There have been a number of attacks on the network recently. For this reason, the network administrator has assigned you the task of configuring Layer 2 security.

For optimum performance and security, the administrator would like to ensure that the root bridge is the 3560 Central switch. To prevent spanning-tree manipulation attacks, the administrator wants to ensure that the STP parameters are secure. To prevent against CAM table overflow attacks, the network administrator has decided to configure port security to limit the number of MAC addresses each switch port can learn. If the number of MAC addresses exceeds the set limit, the administrator would like the port to be shutdown.

All switch devices have been preconfigured with the following:

- Enable password: **ciscoenpa55**
- Console password: **ciscoconpa55**
- SSH username and password: **SSHadmin / ciscosshpa55**

Part 1: Configure Root Bridge

Time Elapsed: 01:01:12 Completion: 76%

Top Dock Check Results Back 1/1 Next

Cisco Packet Tracer - C:\Users\samruddhi thasal\Downloads\SiC All Packet Tracer Files (E-next.in)\p8 Packet Tracer - Layer 2 Security.pka - Guest - 2019-01-02 01:31:34

File Edit Options View Tools Extensions Window Help

Activity Results Time Elapsed: 00:43:51

Congratulations Guest! You completed the activity.

Overall Feedback | Assessment Items | Connectivity Tests

Expand/Collapse All | Show Incorrect Items

Assessment Items	Status	Points	Component(s)	Feedback
Bpduguard	Correct	1	Switching	
Port Security				
Maximum Static MACs	Correct	1	Other	
Port Security Violation	Correct	1	Other	
Sticky Enabled	Correct	1	Other	
PortFast	Correct	1	Switching	
FastEthernet0/3				
Bpduguard	Correct	1	Switching	
Port Security				
Maximum Static MACs	Correct	1	Other	
Port Security Violation	Correct	1	Other	
Sticky Enabled	Correct	1	Other	
PortFast	Correct	1	Switching	
FastEthernet0/4				
Bpduguard	Correct	1	Switching	
Port Security				
Maximum Static MACs	Correct	1	Other	
Port Security Violation	Correct	1	Other	
Sticky Enabled	Correct	1	Other	
PortFast	Correct	1	Switching	
FastEthernet0/5		0	Other	
Port Status	Correct	1	Physical	
FastEthernet0/6		0	Other	
Port Status	Correct	1	Physical	
SW-B				
Ports				
FastEthernet0/1				
Bpduguard	Correct	1	Switching	
Port Security				
Maximum Static MACs	Correct	1	Other	

Score : 50/50
Item Count : 50/50

Component	Items/Total	Score
Other	26/26	26/26
Physical	4/4	4/4
Switching	20/20	20/20

Close

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